

■ Nexus BIM Automation Hub

Project Implementation & Verification Checklist

1. Core Root Configuration

- [x] app.py – Main entry point, session state initialization, and page navigation routing.
- [x] database.py – Database connection wrappers and table initialization logic.
- [x] config.py – Global system variables, base path configurations, and media directory declarations.
- [x] requirements.txt – Application dependencies (streamlit, pandas, plotly, fpdf2, etc.).
- [x] README.md – Project documentation, features, setup guide, and execution commands.
- [x] .gitignore – Rules for ignoring __pycache__, local database files, logs, and uploads.

2. UI Shell & Shared Components (components/)

- [x] login.py – User authentication page and session handler.
- [x] navbar.py – Top header bar with branding and user profile logout controls.
- [x] sidebar.py – Navigation drawer and links.
- [x] footer.py – Standardized enterprise footer.
- [x] dashboard.py – Executive dashboard with system-wide analytics.
- [x] buildings.py – Facilities inventory and building registration form.
- [x] floors.py – Floor layout and spatial breakdown.
- [x] structural.py – Structural element health telemetry.
- [x] models.py – BIM model upload, validation, and storage tracking.
- [x] reports.py – Compliance report renderer and export handlers.
- [x] ai_assistant.py – Interactive chat assistant for natural language BIM queries.
- [x] settings.py – System preferences and integration controls.
- [x] cards.py – Reusable enterprise metric cards.
- [x] charts.py – Custom Plotly chart renderers for health indices and storage pie charts.
- [x] statistics.py – Key performance metrics widget.
- [x] notifications.py – Real-time risk alerts and notifications stream.
- [x] activity.py – Audit trail and user activity log viewer.
- [x] upload.py – Reusable file uploader pipeline component.
- [x] tables.py – Styled data table component wrappers.
- [x] filters.py – Target dropdown filters.
- [x] search.py – Search bar component for asset lookups.
- [x] project_timeline.py – Milestone gantt timeline tracker.
- [x] weather.py – Jobsite environmental telemetry widget.
- [x] calendar_widget.py – Scheduled inspection calendar widget.
- [x] export.py – Data export helper utility.
- [x] utils.py – System logger and file size formatting helpers.

3. Business Logic & Data Domain (modules/)

- [x] users.py – User management and user fetching utilities.
- [x] buildings.py – Facility CRUD operations.
- [x] floors.py – Level allocation logic.
- [x] rooms.py – Room-level space management logic.
- [x] bim_models.py – Model metadata retrieval and storage management.

- [x] structural.py – Structural integrity risk processing.
- [x] equipment.py – Facility MEP & equipment registry logic.
- [x] maintenance.py – Work order tickets and maintenance tasks.
- [x] inspections.py – Audit inspection logs.
- [x] reports.py – Dataset aggregation for compliance exports.
- [x] analytics.py – Risk category categorization logic.
- [x] settings.py – System preferences handler.
- [x] permissions.py – Role-Based Access Control (RBAC) guard logic.

4. Multi-Page Structure (pages/)

- [x] Dashboard.py – Executive Overview Page
- [x] Buildings.py – Facility Inventory Page
- [x] Floors.py – Floor Layout Page
- [x] Rooms.py – Room Registry Page
- [x] BIM_Models.py – BIM Asset Storage Page
- [x] Structural.py – Structural Integrity Monitoring Page
- [x] Reports.py – Analytics & Reports Export Page
- [x] Analytics.py – Telemetry Analytics Page
- [x] AI_Assistant.py – AI Architectural Assistant Page
- [x] Notifications.py – System Alerts Page
- [x] Settings.py – App Settings Page
- [x] Profile.py – User Profile Page

5. Database & Testing Suite

- [x] database/init_db.py – SQLite schema creation and initial seed data generation.
- [x] database/queries.py – SQL helper functions.
- [x] tests/test_login.py – Authentication unit test script.
- [x] tests/test_database.py – Database connection test script.
- [x] tests/test_dashboard.py – Dashboard metric calculation test script.
- [x] tests/test_reports.py – CSV report export test script.